International Journal of Metallurgical & Materials Science and Engineering (IJMMSE)
ISSN 2278-2516
Vol. 3, Issue 2, Jun 2013, 33-38

CRYSTAL GROWTH AND CHARACTERIZATION OF A NEW SEMI ORGANIC NON-LINEAR OPTICAL UREA MAGNESIUM SULPHATE SINGLE CRYSTALS BY SOLUTION GROWTH SLOW EVAPORATION METHOD

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ABSTRACT

Urea Magnesium Sulphate (UMS), a novel semi organic non linear optical crystal has been synthesized using solution growth slow evaporation technique. The lattice parameters for the grown crystals were determined using single crystal XRD. The presence of functional groups for the grown crystals was confirmed using Fourier Transform Infrared (FT-IR) spectroscopy. The optical absorption studies show that the material has wide optical transparency in the entire visible region. The UV cut-off wavelength is found to be at 240 nm. The dielectric constant and dielectric loss has been studied as a function of frequency for various temperatures and the results were discussed in detail. The thermal stability of the crystal was determined from thermo gravimetric and differential thermal analysis curve. The powder sample with average particle size 100 - 115μ were illuminated using Q -switched Nd: YAG laser emitting a fundamental wavelength of 1064 nm with the pulse width of 8ns. The second harmonic generations were confirmed by the emission of green radiation (532 nm). The output power is found to be 1.4 times more than that of KDP.

KEYWORDS: Crystal Growth, Nonlinear Optical Materials, Solution Growth, X-Ray Diffraction

INTRODUCTION

In recent years, second order nonlinear optical materials have attracted many researchers because of their potential applications in various emerging technological fields [1 - 7]. Today, crystal growth technology has advanced rapidly for the development of novel nonlinear optical materials (NLO) for various applications such as optical switching, frequency conversion and electro-optical modulation [8-13]. The organic NLO materials have large nonlinear optical coefficients compared to inorganic material, but their use is impeded by their poor mechanical and thermal properties and low laser damage threshold [14]. The inorganic NLO materials have excellent mechanical and thermal properties but possess relatively modest optical nonlinearities due to lack of extended π-electron delocalization [15]. In view of these problems, a new class of materials has been developed from organic and inorganic complexes called semi organic [16]. In these materials, high optical nonlinearity of pure organic compound is combined with the favorable mechanical and thermal properties of inorganic materials [17-18]. Semi organic crystals have large damage threshold, wide transparency range, less deliquescence, excellent nonlinear optical coefficient, low angular sensitivity and exceptional mechanical properties [19]. Urea has shown interesting properties for nonlinear optical applications [20]. In this present work, single crystals of urea-magnesium sulphate [UMS], a novel nonlinear optical material, has been grown by slow evaporation technique. The grown crystals were characterized by single crystal X-ray diffraction, FTIR, Optical absorption studies and dielectric studies. The SHG efficiency of the grown crystal was measured by Kurtz powder Technique.

SYNTHESIS AND CRYSTAL GROWTH TECHNIQUE

Single crystals of UMS were grown by dissolving (AR Grade) Urea and Magnesium Sulphate in the ratio 1: 1 using triple distilled water. Extreme care was taken to minimize the thermal and mechanical disturbances to the supersaturated solution. After continuous recrystallisation and filtration, optically good quality single crystal having dimensions 25 × 20 × 10 mm^3 is obtained within a period of 9 weeks. The photograph of as grown crystal of UMS is shown in the figure 1.

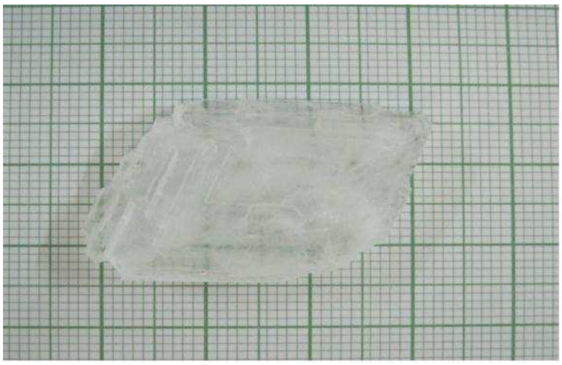

Figure 1: As Grown Good Quality UMS Single Crystals

RESULTS AND DISCUSSIONS

Single Crystal X-Ray Diffraction Analysis

The grown crystals were subjected to single crystal X-ray diffraction analysis using ENRAF NONIUS CAD4 X-ray diffractometer to determine the cell parameters and it reveals that the UMS crystal crystallizes in monoclinic system having non-centrosymmetric space group $P2_1$. The lattice parameters were found to be a = 10.11Å (3), b = 7.21Å (7), c = 24.37Å, and ά=90° β=98.23° γ=90°, V=1758Å

FTIR Analysis

The FTIR spectrum of the grown crystals was recorded in the range 400 – 4000 cm^{-1} using IFS BRUKKER 66V spectrophotometer and the resultant spectrum is shown in the figure 2. The comparison of characteristic vibrational frequencies has been tabulated in table 1.The frequencies at 3372 cm^{-1} is due to symmetric stretching N–H vibration of UMS. The broad band at 2315 cm^{-1} are due to υ_s N=C=O or υ_{as} N=C=O stretching modes. The strong band appearing at 1095 cm^{-1} is due to □ δ symmetric stretching vibration. The band at 630 cm^{-1} is due to C-H mode of vibration. From this FTIR studies shows the characteristic vibration frequencies due to urea and UMS. This confirms the formation of UMS.

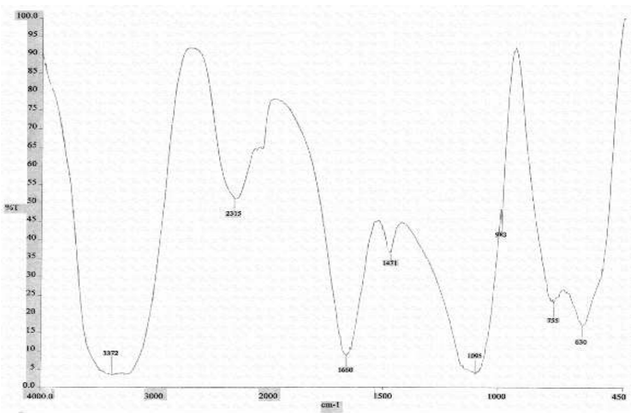

Figure 2: FTIR Spectrum

Table 1: Vibrational Frequenues Assignment on UMS Crystals

Urea	UMS	Assignments
525		δ (N-C-S)
	630	ρ (C-H)
790	755	ν_s (C=S)
1162	983	ν_{as} (C=S)
1008	1095	δ (NH_2)
1454	1471	ν (N-C-N)
1631	1600	δ(NH_2)
	2315	ν_s N=C=O/ν_{as} N=C=O
3320	3372	ν_s(NH_2)
3422		ν_{as}(NH_2)

UV- Vis Spectral Analysis

The optical absorption spectrum for the grown crystals was recorded in the range 200 – 2500 nm using VARIAN CARY 5E SECTROPHOTOMETER and is shown in the figure 3. The resultant spectrum shows that the crystal has very low absorbance in the entire visible and IR region. The UV cut-off wavelength is found to be at 240 nm. This very low absorption property of the grown crystal in the entire visible region suggests its suitability for second harmonic generation [21, 22].

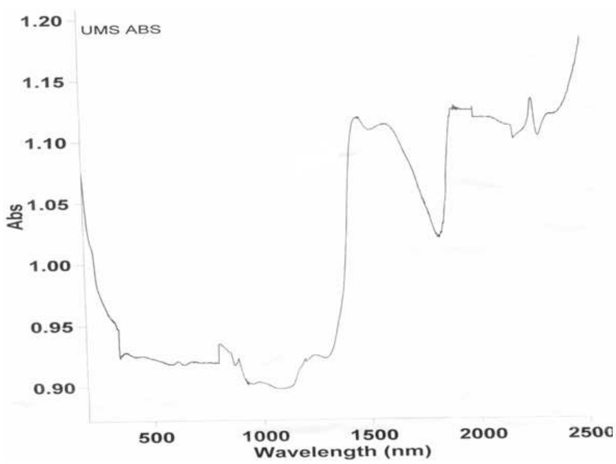

Figure 3: UV–Vis–NIR Absorption Spectrum of UMS

Dielectric Studies

An optically good quality single crystal of UMS was selected for dielectric measurements using HIOKI 3532-50 LCR HITESTER. The selected samples were cut using a diamond saw and polished using paraffin oil. Silver paint was applied on the both faces to make a capacitor with the crystal as a dielectric material. The dielectric constant is calculated using the relation

$$\varepsilon' = Cd / \varepsilon_0 A$$

Where C is the capacitance, d is the thickness, A is the area and ε_0 is the absolute permittivity of the free space (8.854×10^{-12} F/m). The variation of dielectric constant (ε’) was studied as a function of frequency for the grown crystal at various temperatures viz., 388, 333,348 and 363 K and is shown in Figure 4. The high value of dielectric constant at low frequencies may be due to the presence of all the four polarizations and its low value at higher frequencies may be due to the loss of significance of these polarizations gradually [23].

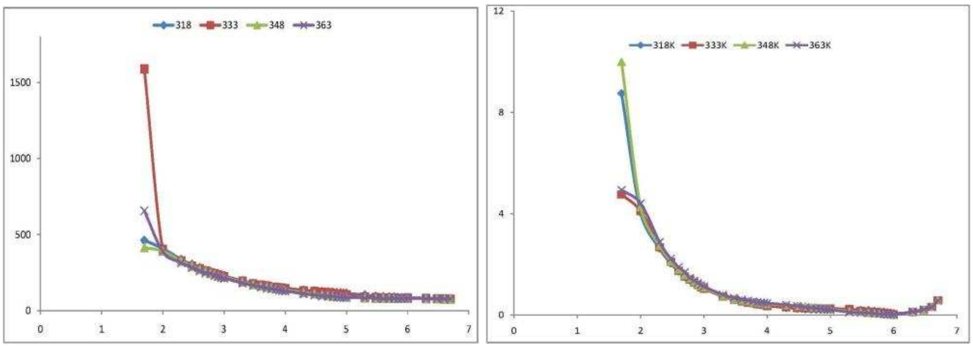

Figure 4: Dielectric Constant Versus Log F and Dielectric Loss Versus Log F

From the plot, it is also observed that dielectric constant decreases with increase in frequency. The variation of dielectric loss with frequency is shown in Figure 4. The characteristics of low dielectric loss at very high frequency suggest that it possesses enhanced optical quality with lesser defects and this parameter is essential for nonlinear optical applications [24].

Thermal Analysis

The thermo gravimetric analysis of urea – magnesium sulphate crystals was carried out for the sample weight of 8.820 mg between 30 to 830˚ C at a heating rate of 20 K min^{-1} in nitrogen atmosphere using NETZSCH STA 409 C/CD thermal analyzer and the resultant spectrum is shown in the figure 5. The TGA illustrates that there is no loss below 180º C illustrating the absence of water in the crystal lattice and the sharp weight loss at 180º C, without any intermediate stages, is assigned as melting point of the crystal. DTA studies also reveal that above 180˚C, the material begins to attain an endothermic transition and starts to decompose at 225˚C. The DTA thermo gram confirms that the endothermic peaks coincides with that of TGA, thus confirms the thermal stability of the crystal. The sharpness of this peak shows the good degree of crystallinity and purity of the sample. Thus from the thermal studies, the crystal can retain its texture upto 180º C.

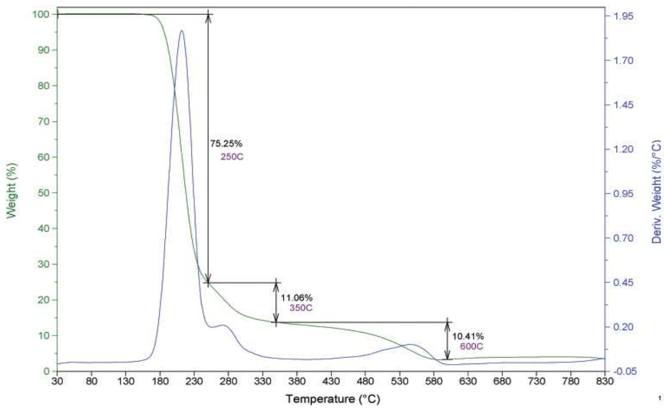

Figure 5: TG/DTA Spectrum of the UMS Crystal

Kurtz Powder (NLO) Test

The second harmonic generation efficiency measurement was carried out on the grown crystal using the Kurtz–Perry powder technique. The crystal was grounded into a homogenous powder of particles and densely packed between two transparent glass slides. The powder sample with average particle size 100 - 115µ were illuminated using Q -switched Nd: YAG laser emitting a fundamental wavelength of 1064 nm with the pulse width of 8ns. The second harmonic generations were confirmed by the emission of green radiation (532 nm). The output power is found to be 1.4 times more than that of KDP.

CONCLUSIONS

A new semi organic NLO material UMS has been synthesized and crystals were grown by slow evaporation methods. The lattice parameter values have been evaluated by single- crystal XRD analysis. The UV-Vis spectra reveals that the crystals are transparent in the entire visible region with cut-off wavelength at 240nm. Which is an eager property for NLO application. FTIR analysis confirms the presence of functional groups. Dielectric constant decreases with increase in frequency and very low values of dielectric loss infers very high purity of the crystal. The TGA and DTA thermogram predicts the stability of the material decomposition and corresponding weight losses. The emission of green radiation confirms the second harmonic generation

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